Herschel Pravin Pawar

Software Engineer specializing in high-performance Rust systems programming and safe concurrent data pipelines. Focused on minimizing barriers in multi-threading, CPU cache locality, and parallel computing architectures. Committed to technical mentoring, building inclusive engineering cultures, and writing deep-dive technical documentation.

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Visit My Website

Education

Bachelors of Engineering in Information Technology

2020 - 2024

University of Mumbai

MGM College of Engineering and Technology, Kamothe, Navi Mumbai, India

Projects & Associations

Personal Website

April 2024 - Present

Digital Garden

https://sakurakat.systems/posts/

- A static blog built with Astro that hosts technical posts, tutorials, and write-ups.
- Integrated a Typst processing plugin into the Astro site layout to enable parallel, backwards-compatible authoring while leveraging custom templates for deterministic social preview graphic generation.

Astro | Typst | Blog | Rust | Nix | Web | Game Development

Parrylord

June 2025

Bevy Jam 6

https://github.com/pawarherschel/parrylord

- Utilized the Bevy Entity Component System (ECS) paradigm to decouple game execution logic from state data, leveraging data-driven composition over inheritance to manage real-time event loops.
- Engineered a serverless global leaderboard system utilizing Cloudflare Workers and D1 SQL databases.

Game Development | Bevy | Rust | ECS

BF Compiler with Cranelift

November 2024

Personal Research

https://github.com/pawarherschel/BFCranelift

- Utilized technical blogs to build a compiler emitting native x86_64 machine code from raw instructions.
- Integrated the Capstone disassembly engine to dump instruction layouts for manual validation and debugging of the generated assembly.

BF | Compiler | Cranelift | Rust

VRCX Insights

2023 - 2024

Personal Project

https://github.com/pawarherschel/vrcx-insights

- Designed an asynchronous pipeline to handle non-blocking relational database lookups concurrently across 15 worker threads without data races.
- Modeled relational network data into a graph topology, exporting dot graphs for visual exploration.

Data Mining | Rust | SQLite | SQLx | tokio

Declarative NixOS VM Deployment on Hyper-V

April 2025 - May 2025

Personal Research

https://sakurakat.systems/posts/hyperv-shenanigans/

- Engineered a reproducible pipeline to cross-compile and deploy custom NixOS images directly onto an isolated Windows Hyper-V fabric.
- Documented hypervisor constraints to mitigate VHDX errors and secure boot blocks, while configuring local profiles with hashed credentials for TTY authentication boundaries.

NixOS | Virtualization | Hyper-V | Nix Flakes

Skills

Programming Languages	Rust | GDScript | TypeScript | Nixlang | C++ | C#
Systems & Tooling	Nix | Typst | Profiling | Git
Familiar Areas	Docker (Containerization) | Cloudflare (Hosting, Workers, D1, R2, etc) | Web Scraping | Astro (Web Frontend)